

PI/PID Controller Practice Questions with Answers

Includes 5-step numerical updates, overshoot analysis, response plots, and Ziegler-Nichols tuning questions using the slide formulas

General Instructions

For each numerical controller question, compute five update steps and plot the output against time. Use the reference value as a horizontal line in the graph. Unless stated otherwise, take sampling time $\Delta t = 1$ s.

Question 1: Pendulum Regulation using PI Controller

A simple pendulum is displaced by an angle θ from the vertical. The goal is to regulate the pendulum to $\theta_{ref} = 0$ degree using a PI controller.

- Initial value: $\theta(0) = 40$ degree
- Reference: $\theta_{ref} = 0$ degree
- Sampling time: $\Delta t = 1$ s
- Controller gains: $K_p = 0.5$, $K_i = 0.1$
- Error: $e_k = \theta_{ref} - \theta_k$
- Cumulative error: $S_k = \text{sum of errors from } i = 0 \text{ to } k$
- Control: $u_k = K_p e_k + K_i S_k$
- System output: $\theta_{k+1} = \theta_k + u_k \Delta t$

Tasks:

1. Compute the next 5 steps: $k = 0, 1, 2, 3, 4$.
2. Fill the table below.
3. Plot θ_k versus time step k and draw the reference line.
4. State whether overshoot occurs. If yes, mention the first step and explain why.

k	θ_k	e_k	S_k	u_k	θ_{k+1}

Question 2: DC Motor Speed Control using PI Controller

A DC motor is rotating below the desired speed. The goal is to regulate the motor speed to $\omega_{ref} = 100$ rpm using a PI controller.

- Initial value: $\omega(0) = 20$ rpm
- Reference: $\omega_{ref} = 100$ rpm
- Sampling time: $\Delta t = 1$ s
- Controller gains: $K_p = 0.3$, $K_i = 0.05$
- Error: $e_k = \omega_{ref} - \omega_k$
- Cumulative error: $S_k = \text{sum of errors from } i = 0 \text{ to } k$
- Control: $u_k = K_p e_k + K_i S_k$
- System output: $\omega_{k+1} = \omega_k + u_k \Delta t$

Tasks:

5. Compute the next 5 steps: $k = 0, 1, 2, 3, 4$.
6. Fill the table below.
7. Plot ω_k versus time step k and draw the reference line.
8. State whether overshoot occurs. If yes, mention the first step and explain why.

k	ω_k	e_k	S_k	u_k	ω_{k+1}

Question 3: Mobile Robot Position Control using PI Controller

A mobile robot is moving along a straight line. The goal is to move it to $x_{ref} = 10$ m using a PI controller.

- Initial value: $x(0) = 0$ m
- Reference: $x_{ref} = 10$ m
- Sampling time: $\Delta t = 1$ s
- Controller gains: $K_p = 0.4$, $K_i = 0.08$
- Error: $e_k = x_{ref} - x_k$
- Cumulative error: $S_k = \text{sum of errors from } i = 0 \text{ to } k$
- Control: $u_k = K_p e_k + K_i S_k$
- System output: $x_{k+1} = x_k + u_k \Delta t$

Tasks:

9. Compute the next 5 steps: $k = 0, 1, 2, 3, 4$.
10. Fill the table below.
11. Plot x_k versus time step k and draw the reference line.
12. State whether overshoot occurs. If yes, mention the first step and explain why.

k	x_k	e_k	S_k	u_k	x_{k+1}

Question 4: Temperature Control using PID Controller

A heater is used to control room temperature. The goal is to reach $T_{ref} = 30$ degree C using a PID controller.

- Initial value: $T(0) = 20$ degree C
- Reference: $T_{ref} = 30$ degree C
- Sampling time: $\Delta t = 1$ s
- Controller gains: $K_p = 0.6$, $K_i = 0.05$, $K_d = 0.2$
- Error: $e_k = T_{ref} - T_k$
- Cumulative error: $S_k = \text{sum of errors from } i = 0 \text{ to } k$
- Derivative error: $D_k = e_k - e_{k-1}$; assume $e_{-1} = 0$
- Control: $u_k = K_p e_k + K_i S_k + K_d D_k$
- System output: $T_{k+1} = T_k + u_k \Delta t$

Tasks:

13. Compute the next 5 steps: $k = 0, 1, 2, 3, 4$.
14. Fill the table below.
15. Plot T_k versus time step k and draw the reference line.
16. State whether overshoot occurs. If yes, mention the first step and explain why.

k	T_k	e_k	S_k	D_k	u_k	T_{k+1}

Question 5: Drone Altitude Control using PID Controller

A drone is flying below the target altitude. The goal is to reach $h_{ref} = 10$ m using a PID controller.

- Initial value: $h(0) = 2$ m
- Reference: $h_{ref} = 10$ m
- Sampling time: $\Delta t = 1$ s
- Controller gains: $K_p = 0.5$, $K_i = 0.1$, $K_d = 0.15$
- Error: $e_k = h_{ref} - h_k$
- Cumulative error: $S_k = \text{sum of errors from } i = 0 \text{ to } k$
- Derivative error: $D_k = e_k - e_{k-1}$; assume $e_{-1} = 0$
- Control: $u_k = K_p e_k + K_i S_k + K_d D_k$
- System output: $h_{k+1} = h_k + u_k \Delta t$

Tasks:

17. Compute the next 5 steps: $k = 0, 1, 2, 3, 4$.
18. Fill the table below.
19. Plot h_k versus time step k and draw the reference line.
20. State whether overshoot occurs. If yes, mention the first step and explain why.

k	h_k	e_k	S_k	D_k	u_k	h_{k+1}

Ziegler-Nichols PID Tuning Questions (Using Slide Rules)

ZN Question 1: Closed-loop ultimate gain method

During a closed-loop test using proportional control only, a robot heading controller shows sustained oscillation when $K = 8$ and the oscillation period is $P = 4$ s. Using the Ziegler-Nichols rules from the slide, calculate P-only, PI, and PID controller gains.

- Task: Show the formula and calculate the required controller gains.

ZN Question 2: Closed-loop tuning for motor speed

A DC motor speed controller reaches sustained oscillation at gain $K = 12.5$. The measured oscillation period is $P = 2.5$ s. Using the Ziegler-Nichols rules from the slide, calculate the PID gains K_p , K_i , and K_d .

- Task: Show the formula and calculate the required controller gains.

ZN Question 3: Closed-loop tuning for temperature control

A temperature controller is tested using proportional control only. The system produces sustained oscillation at gain $K = 6$ and oscillation period $P = 5$ s. Using the Ziegler-Nichols rules from the slide, calculate P-only, PI, and PID controller gains.

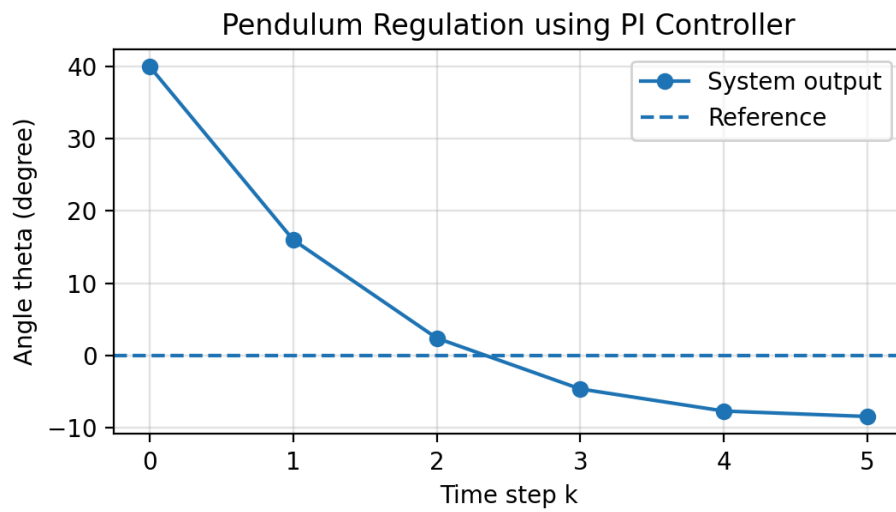
- Task: Show the formula and calculate the required controller gains.

Answer Key

Answer: Question 1: Pendulum Regulation using PI Controller

k	theta k	e k	S k	u k	theta next
0	40	-40	-40	-24	16
1	16	-16	-56	-13.6	2.4
2	2.4	-2.4	-58.4	-7.04	-4.64
3	-4.64	4.64	-53.76	-3.056	-7.696
4	-7.696	7.696	-46.064	-0.758	-8.454

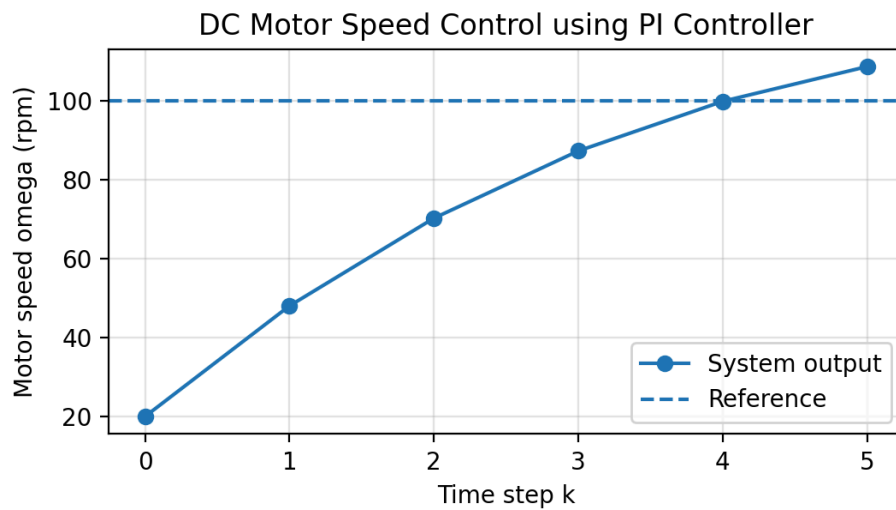
Overshoot: Yes. It first occurs after step $k = 2$, when the next output crosses the reference at time step 3.



Answer: Question 2: DC Motor Speed Control using PI Controller

k	ω_k	e_k	S_k	u_k	ω_{next}
0	20	80	80	28	48
1	48	52	132	22.2	70.2
2	70.2	29.8	161.8	17.03	87.23
3	87.23	12.77	174.57	12.559	99.79
4	99.79	0.21	174.78	8.802	108.592

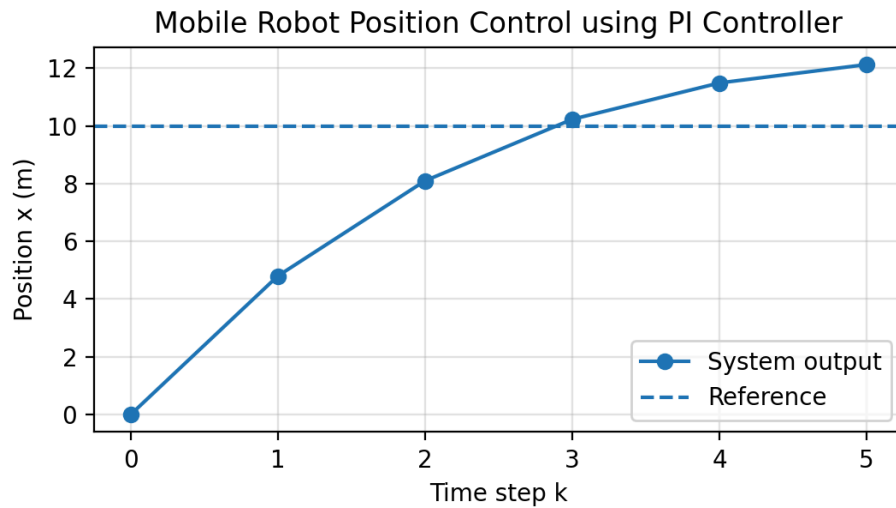
Overshoot: Yes. It first occurs after step $k = 4$, when the next output crosses the reference at time step 5.



Answer: Question 3: Mobile Robot Position Control using PI Controller

k	x k	e k	S k	u k	x next
0	0	10	10	4.8	4.8
1	4.8	5.2	15.2	3.296	8.096
2	8.096	1.904	17.104	2.13	10.226
3	10.226	-0.226	16.878	1.26	11.486
4	11.486	-1.486	15.392	0.637	12.123

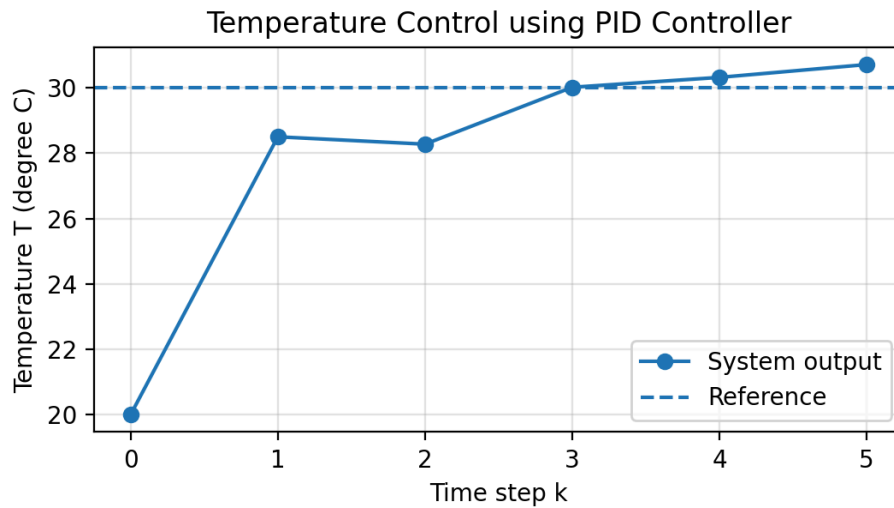
Overshoot: Yes. It first occurs after step $k = 2$, when the next output crosses the reference at time step 3.



Answer: Question 4: Temperature Control using PID Controller

k	T k	e k	S k	D k	u k	T next
0	20	10	10	10	8.5	28.5
1	28.5	1.5	11.5	-8.5	-0.225	28.275
2	28.275	1.725	13.225	0.225	1.741	30.016
3	30.016	-0.016	13.209	-1.741	0.302	30.319
4	30.319	-0.319	12.89	-0.302	0.393	30.711

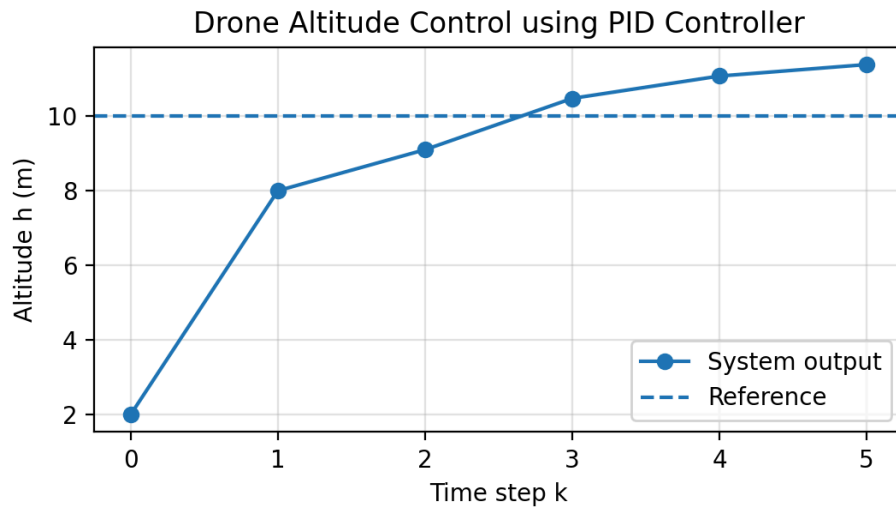
Overshoot: Yes. It first occurs after step $k = 2$, when the next output crosses the reference at time step 3.



Answer: Question 5: Drone Altitude Control using PID Controller

k	h k	e k	S k	D k	u k	h next
0	2	8	8	8	6	8
1	8	2	10	-6	1.1	9.1
2	9.1	0.9	10.9	-1.1	1.375	10.475
3	10.475	-0.475	10.425	-1.375	0.599	11.074
4	11.074	-1.074	9.351	-0.599	0.308	11.382

Overshoot: Yes. It first occurs after step $k = 2$, when the next output crosses the reference at time step 3.



Ziegler-Nichols Answer Key

Answer: ZN Question 1

Slide rules used: P control: $K_p = 0.5K$. PI control: $K_p = 0.45K$, $K_i = 1.2/P$. PID control: $K_p = 0.6K$, $K_i = 2.0/P$, $K_d = P/8$.

For $K = 8$ and $P = 4$ s: P-only $K_p = 0.5(8) = 4$. PI: $K_p = 0.45(8) = 3.6$, $K_i = 1.2/4 = 0.30$. PID: $K_p = 0.6(8) = 4.8$, $K_i = 2.0/4 = 0.50$, $K_d = 4/8 = 0.50$.

Controller	K_p	K_i	K_d
P	4	-	-
PI	3.6	1.08	-
PID	4.8	2.4	2.4

Answer: ZN Question 2

Using the slide PID rules with $K = 12.5$ and $P = 2.5$ s: $K_p = 0.6K = 0.6(12.5) = 7.5$, $K_i = 2.0/P = 2.0/2.5 = 0.80$, $K_d = P/8 = 2.5/8 = 0.3125$.

K_p	K_i	K_d
7.5	6	2.344

Answer: ZN Question 3

Using the slide rules with $K = 6$ and $P = 5$ s: P-only $K_p = 0.5(6) = 3$. PI: $K_p = 0.45(6) = 2.7$, $K_i = 1.2/5 = 0.24$. PID: $K_p = 0.6(6) = 3.6$, $K_i = 2.0/5 = 0.40$, $K_d = 5/8 = 0.625$.

K_p	T_i	T_d	K_i	K_d
4	3	0.75	1.333	3